

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

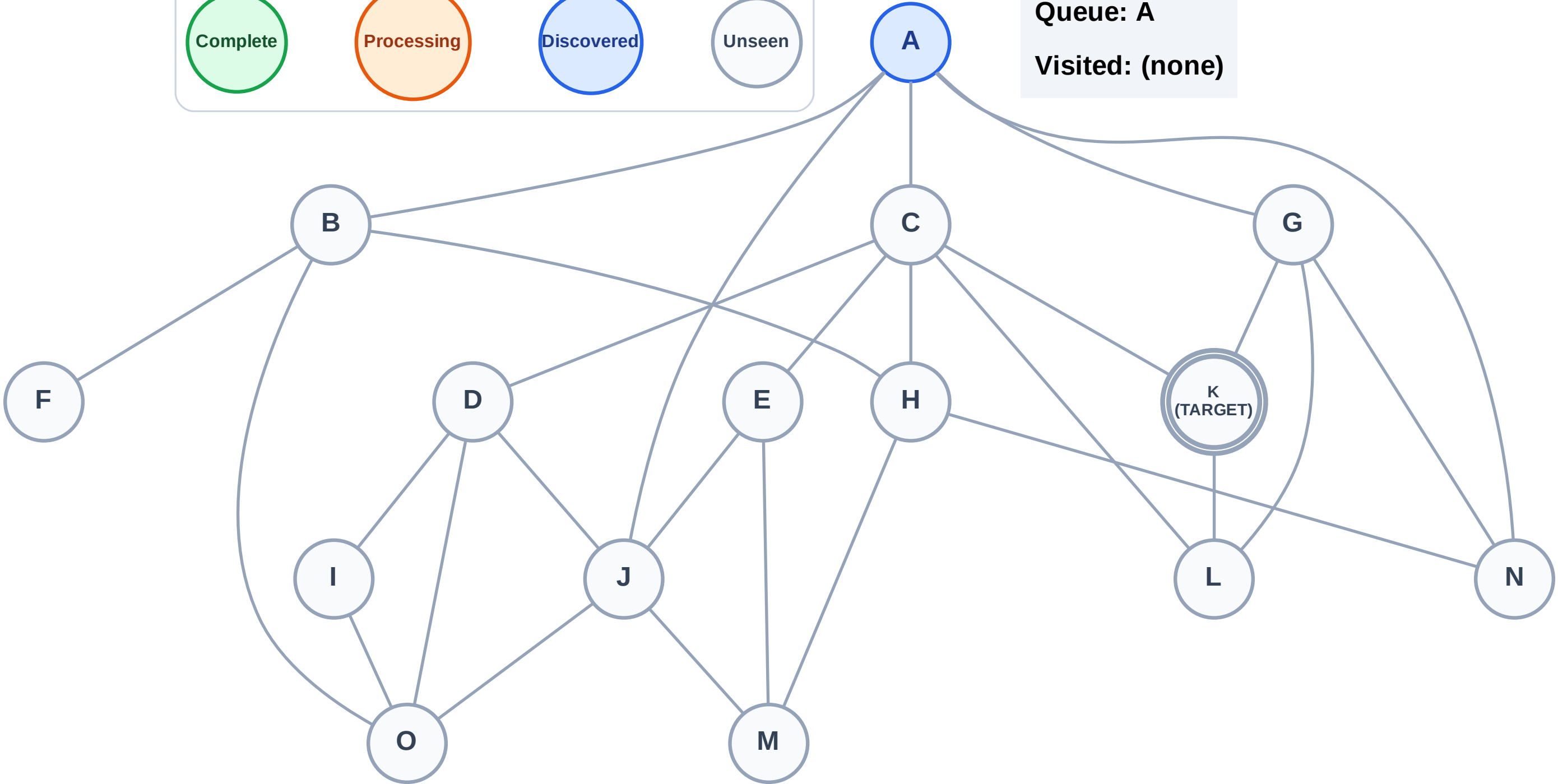
Processing

Discovered

Unseen

Queue: A

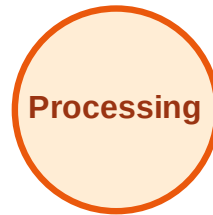
Visited: (none)



BFS Traversal

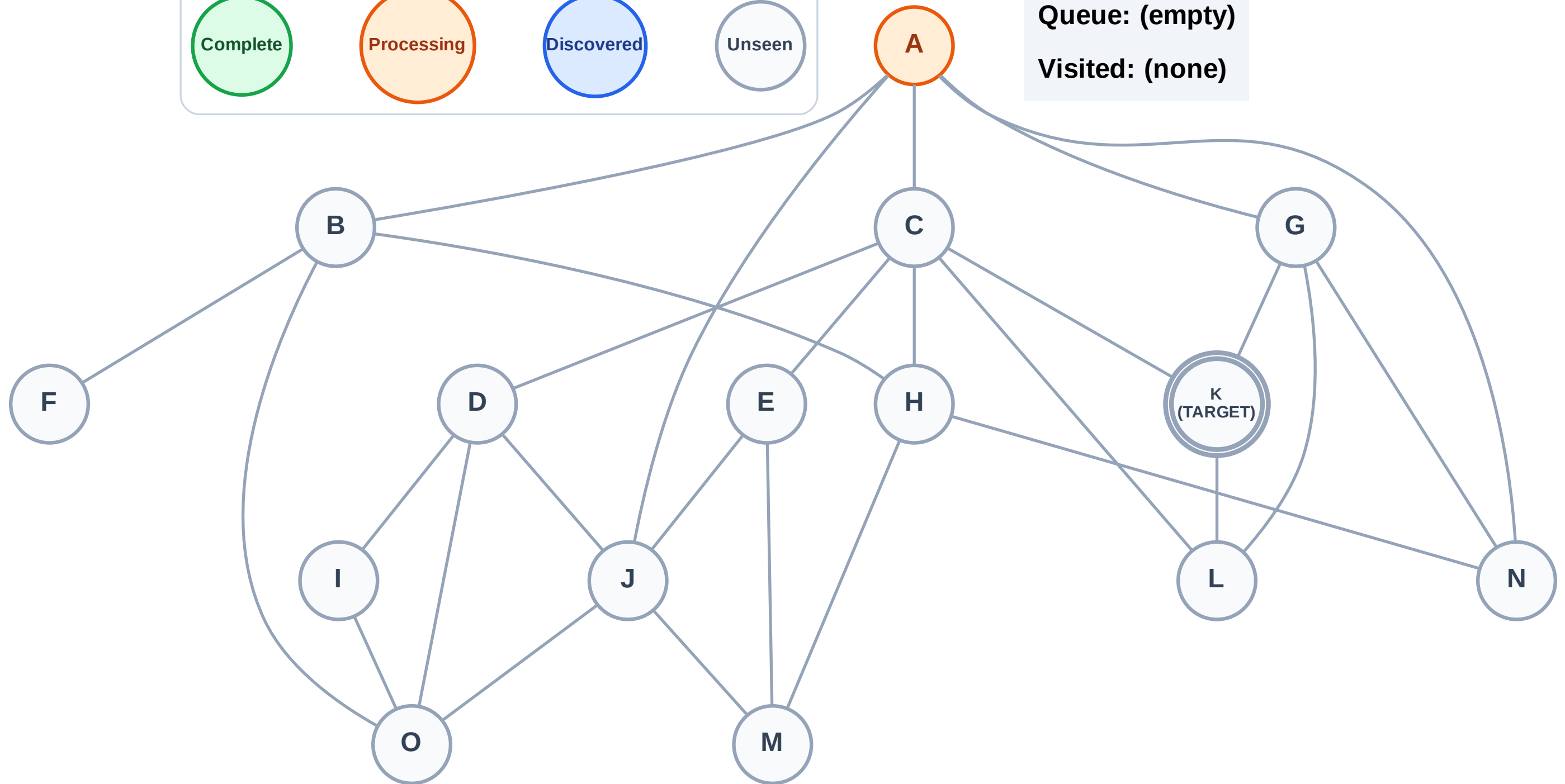
Step 2: Process node A

Legend



Queue: (empty)

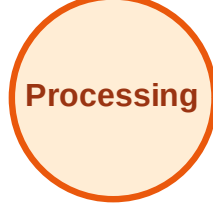
Visited: (none)



BFS Traversal

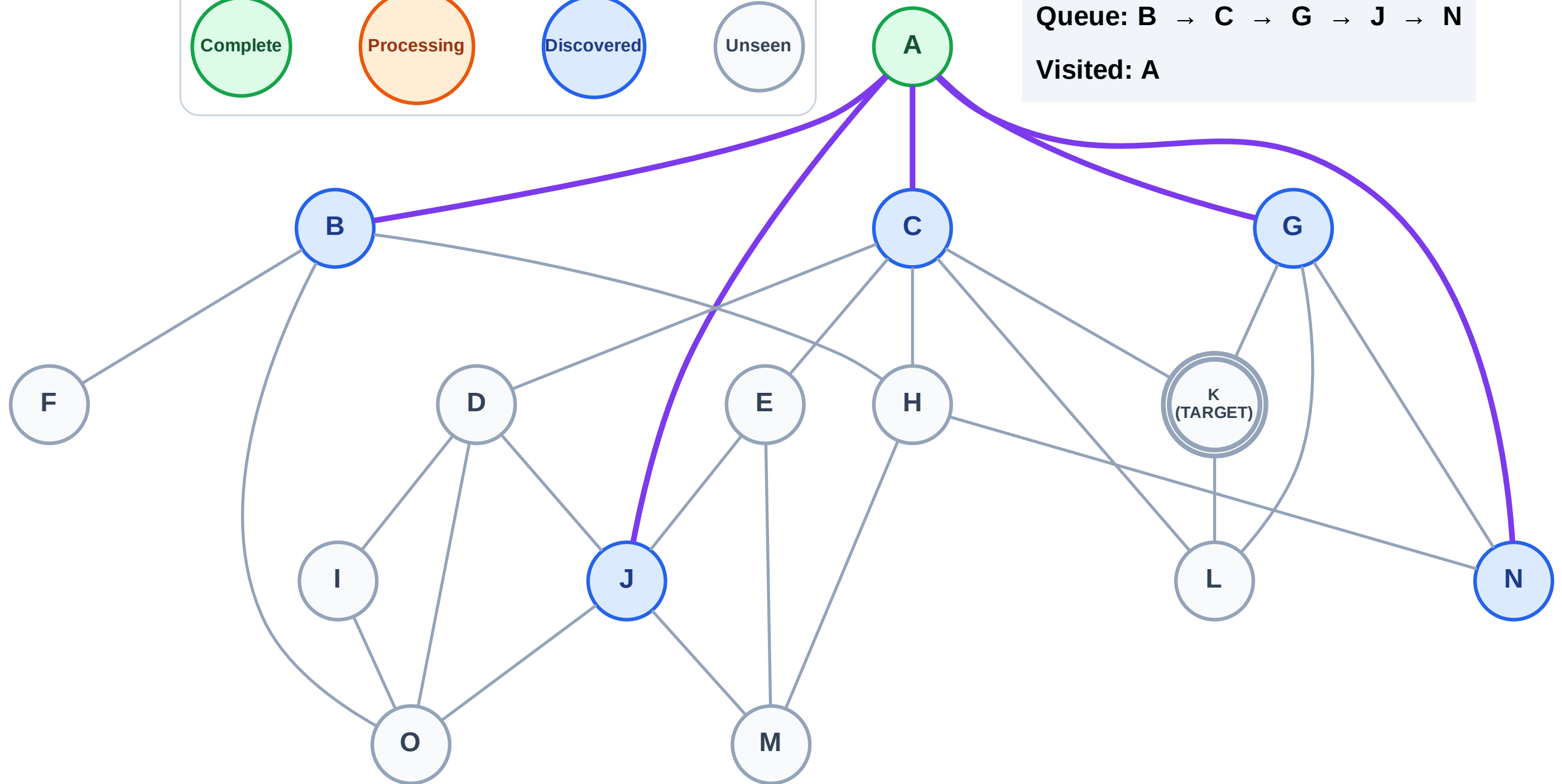
Step 3: Discover B, C, G, J, N from A

Legend



Queue: B → C → G → J → N

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

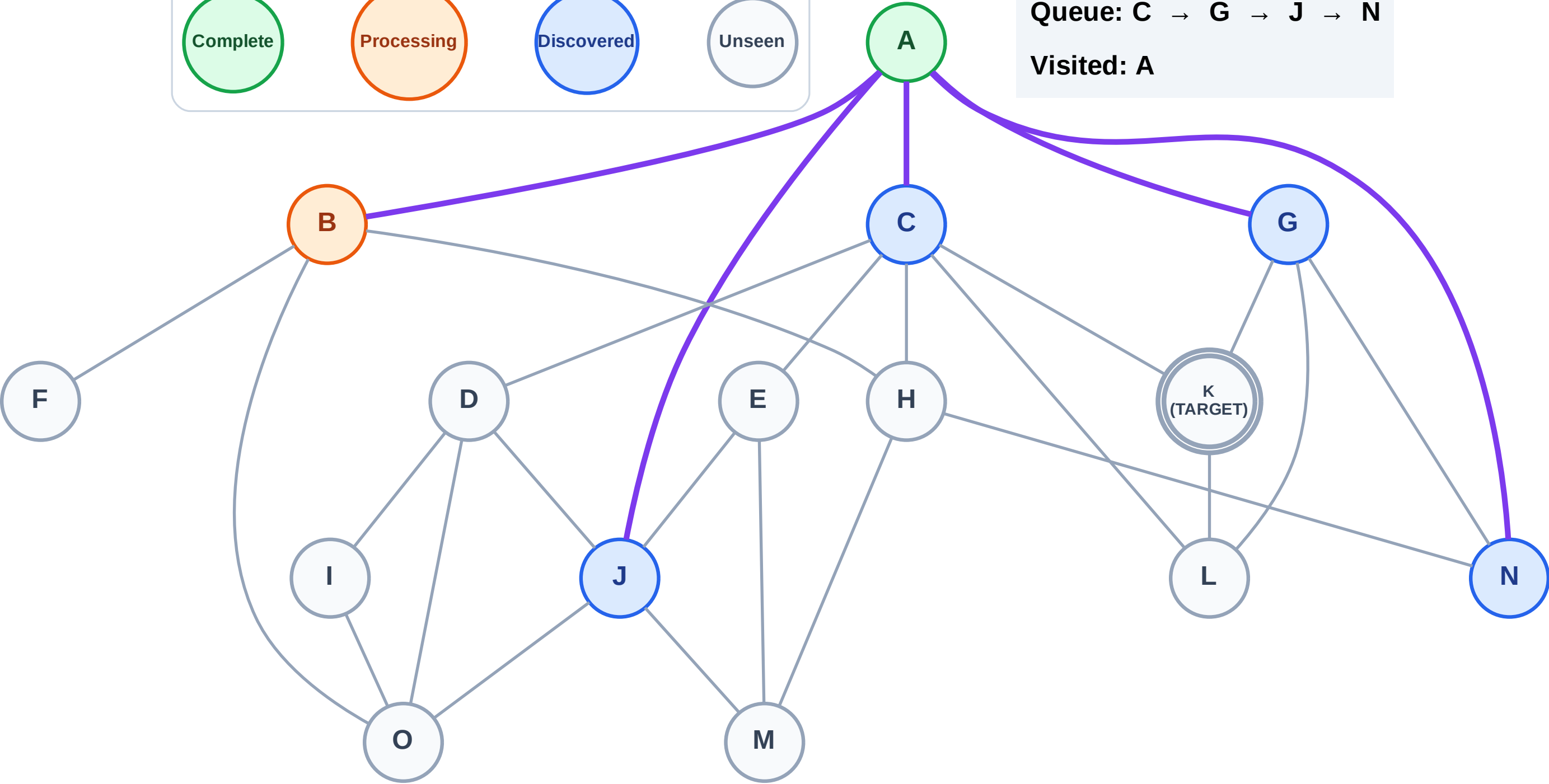
Processing

Discovered

Unseen

Queue: C → G → J → N

Visited: A



BFS Traversal

Step 5: Discover F, H, O from B

Legend

Complete

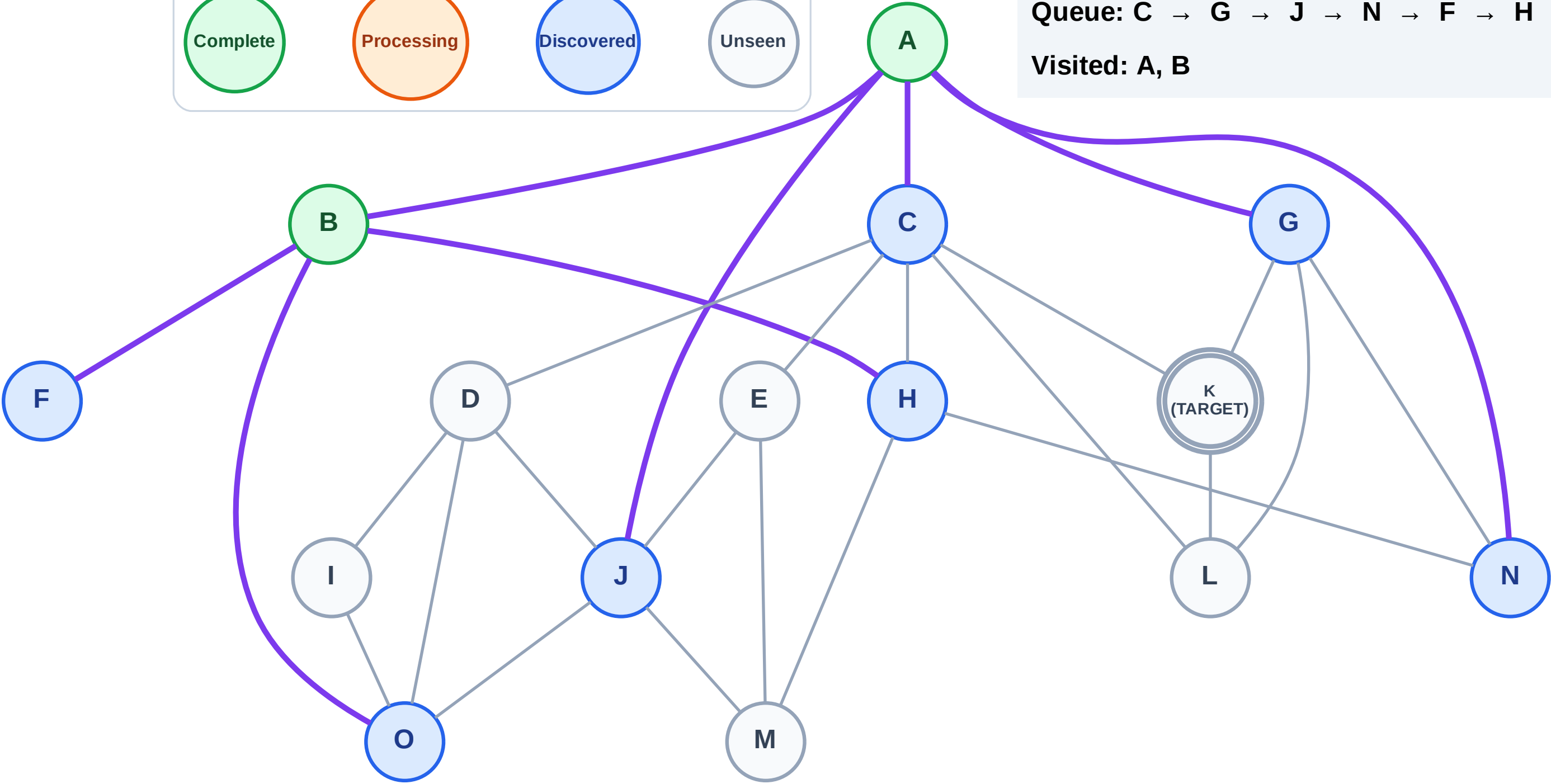
Processing

Discovered

Unseen

Queue: C → G → J → N → F → H → O

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

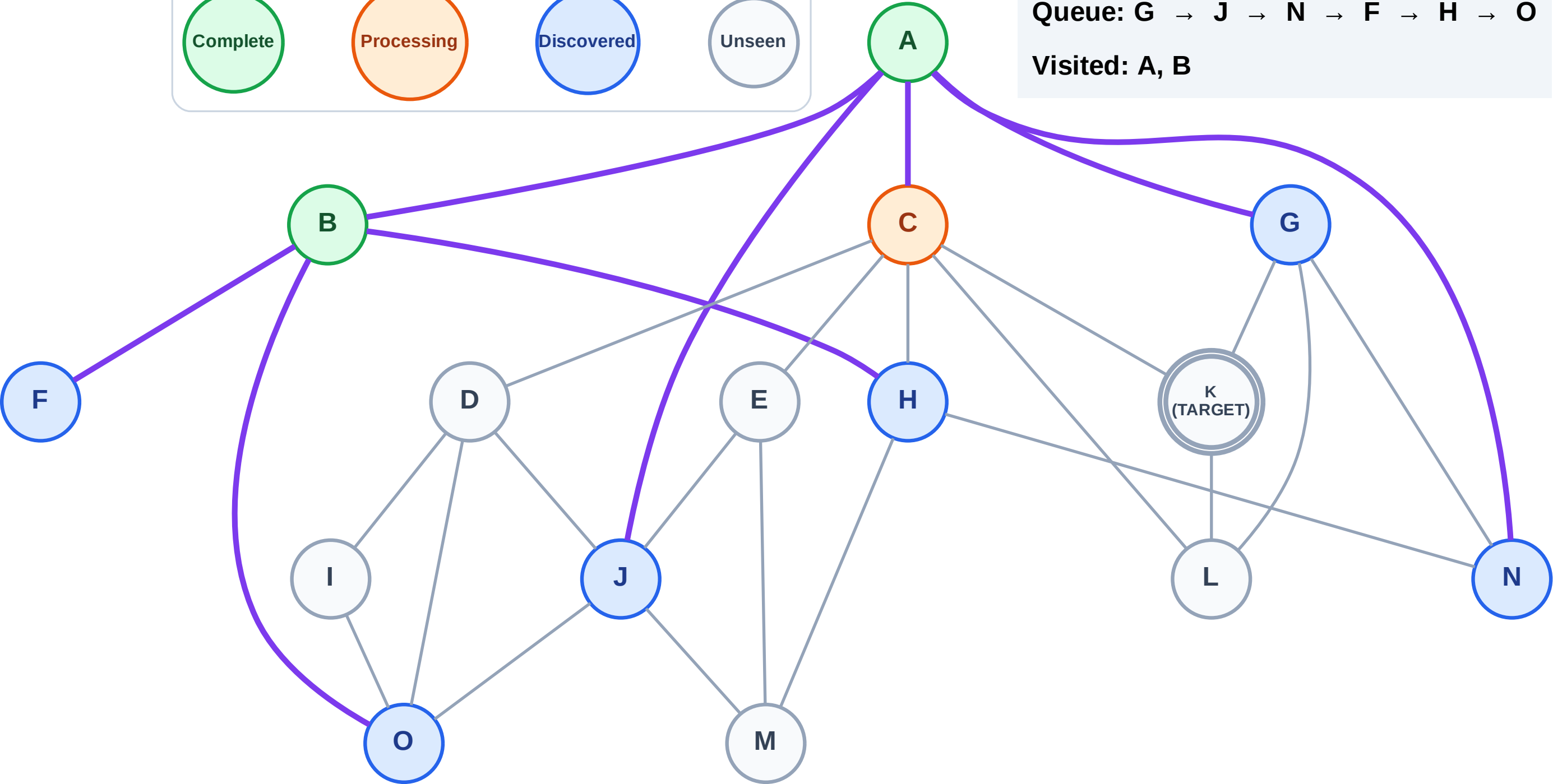
Processing

Discovered

Unseen

Queue: G → J → N → F → H → O

Visited: A, B



BFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → C → K

Legend

Complete

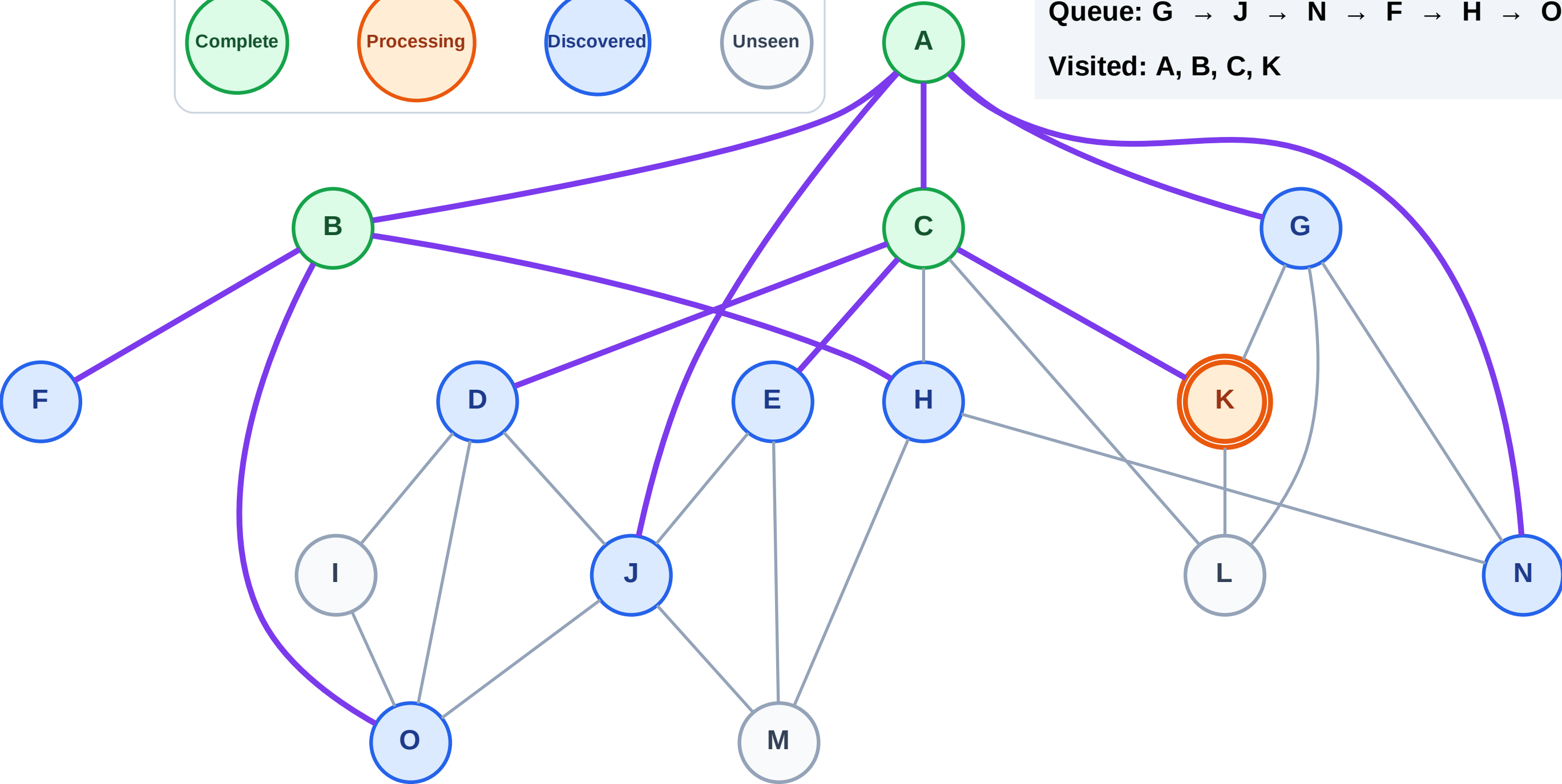
Processing

Discovered

Unseen

Queue: G → J → N → F → H → O → D → E

Visited: A, B, C, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

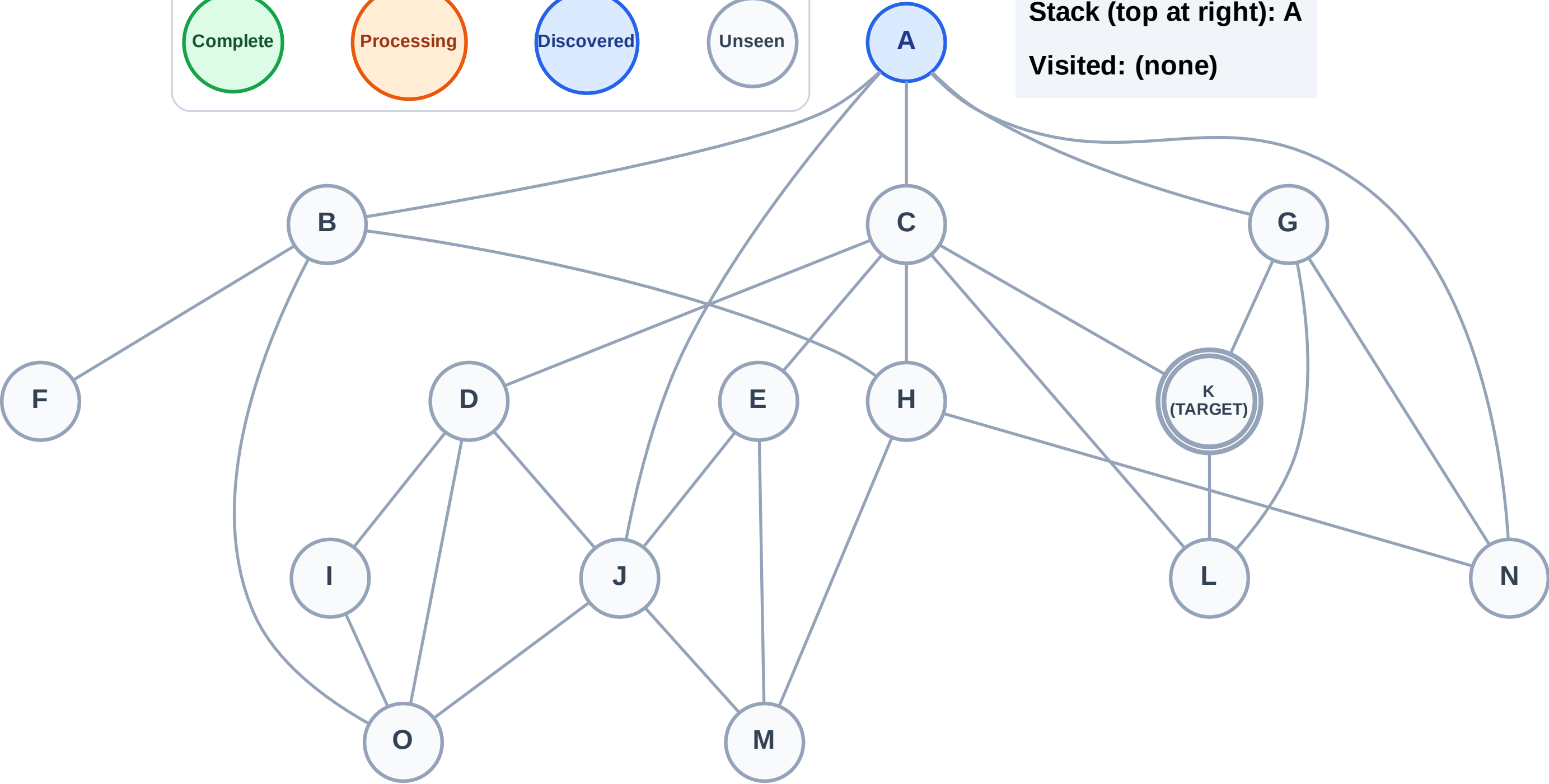
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



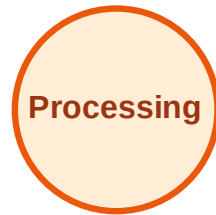
DFS Traversal

Step 2: Process node A

Legend



Complete



Processing



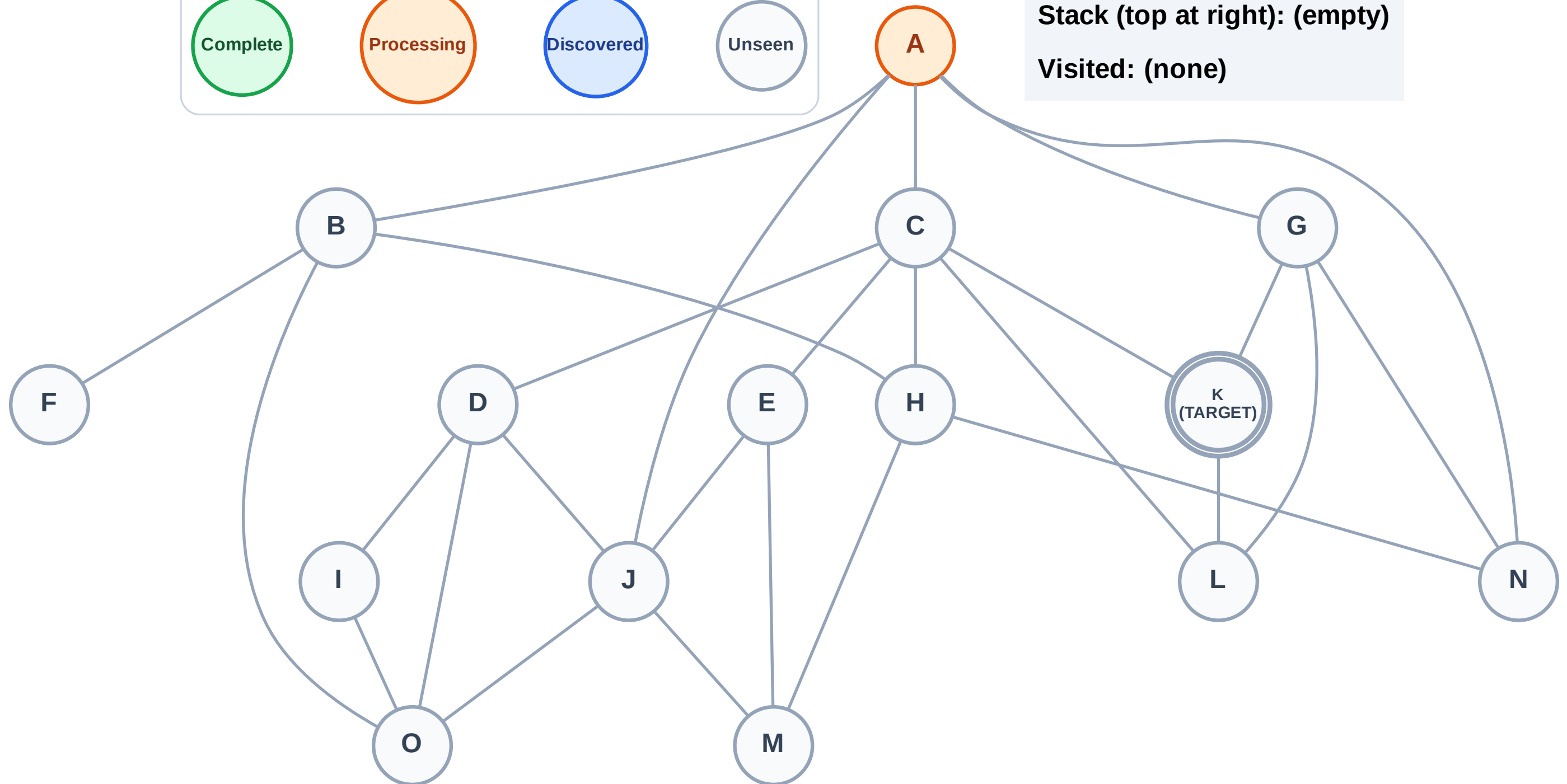
Discovered



Unseen

Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover N, J, G, C, B from A

Legend

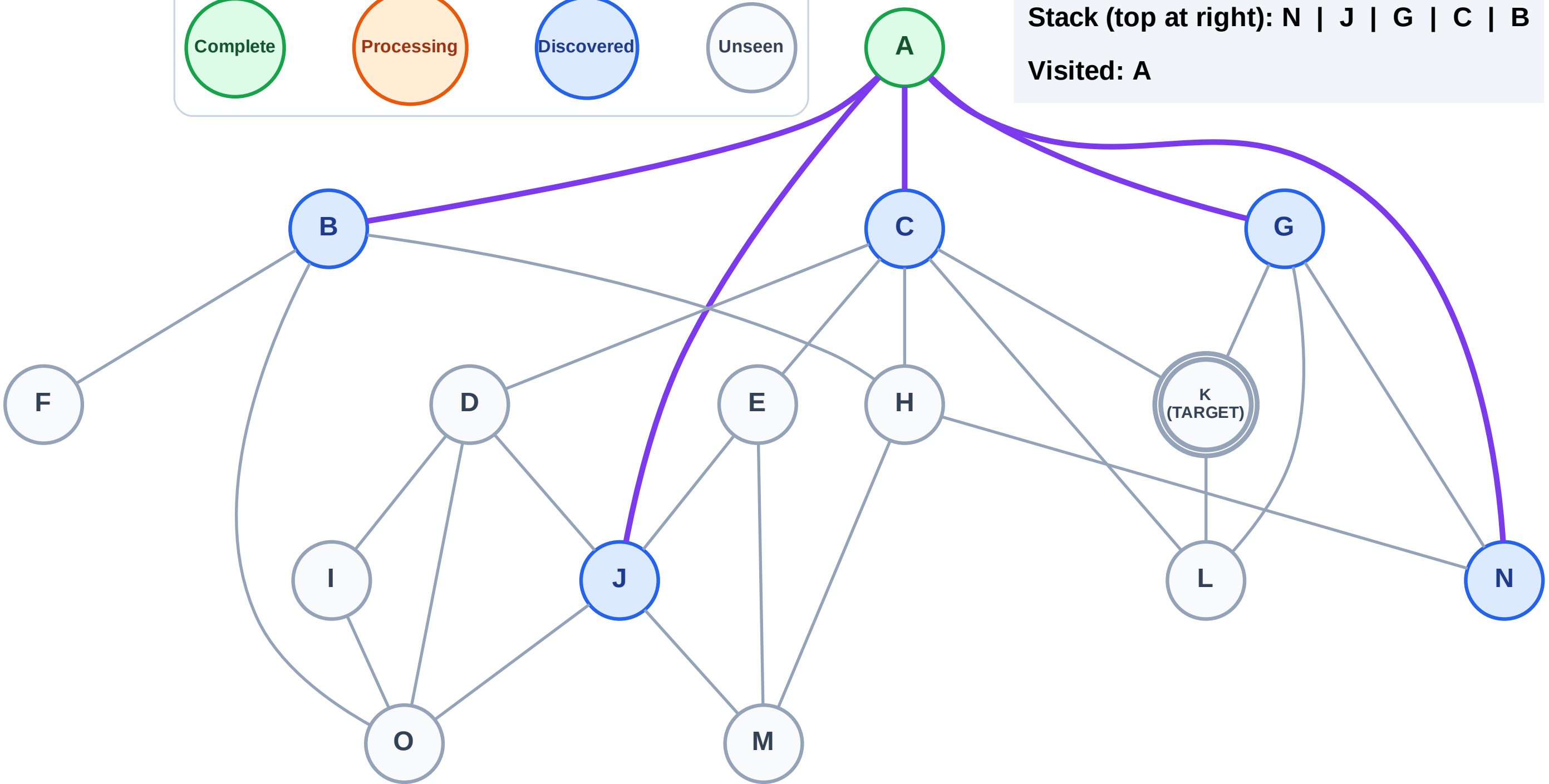
Complete

Processing

Discovered

Unseen

Stack (top at right): N | J | G | C | B
Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

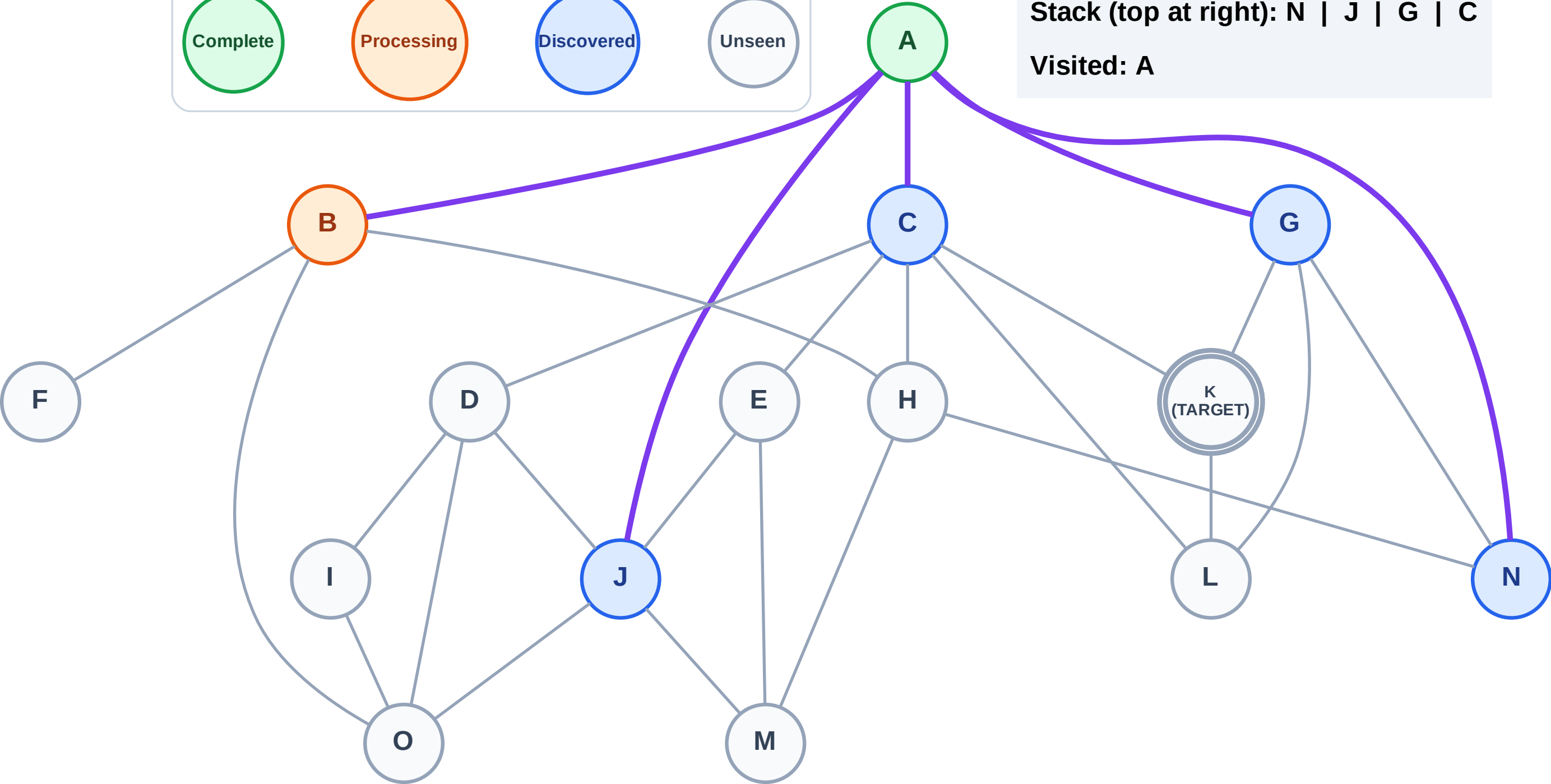
Processing

Discovered

Unseen

Stack (top at right): N | J | G | C

Visited: A



DFS Traversal

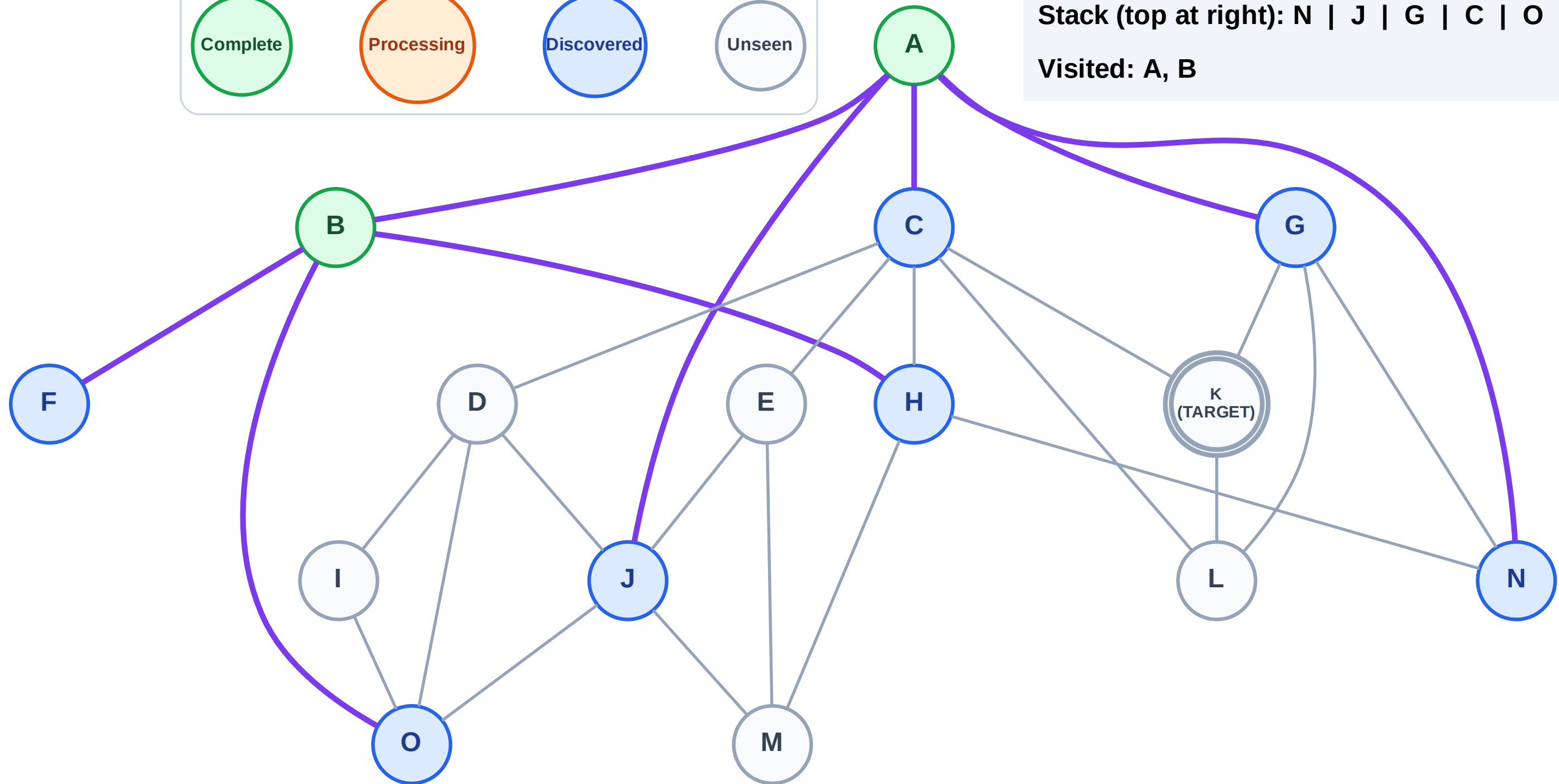
Step 5: Discover O, H, F from B

Legend



Stack (top at right): N | J | G | C | O | H | F

Visited: A, B



DFS Traversal

Step 6: Process node F

Legend

Complete

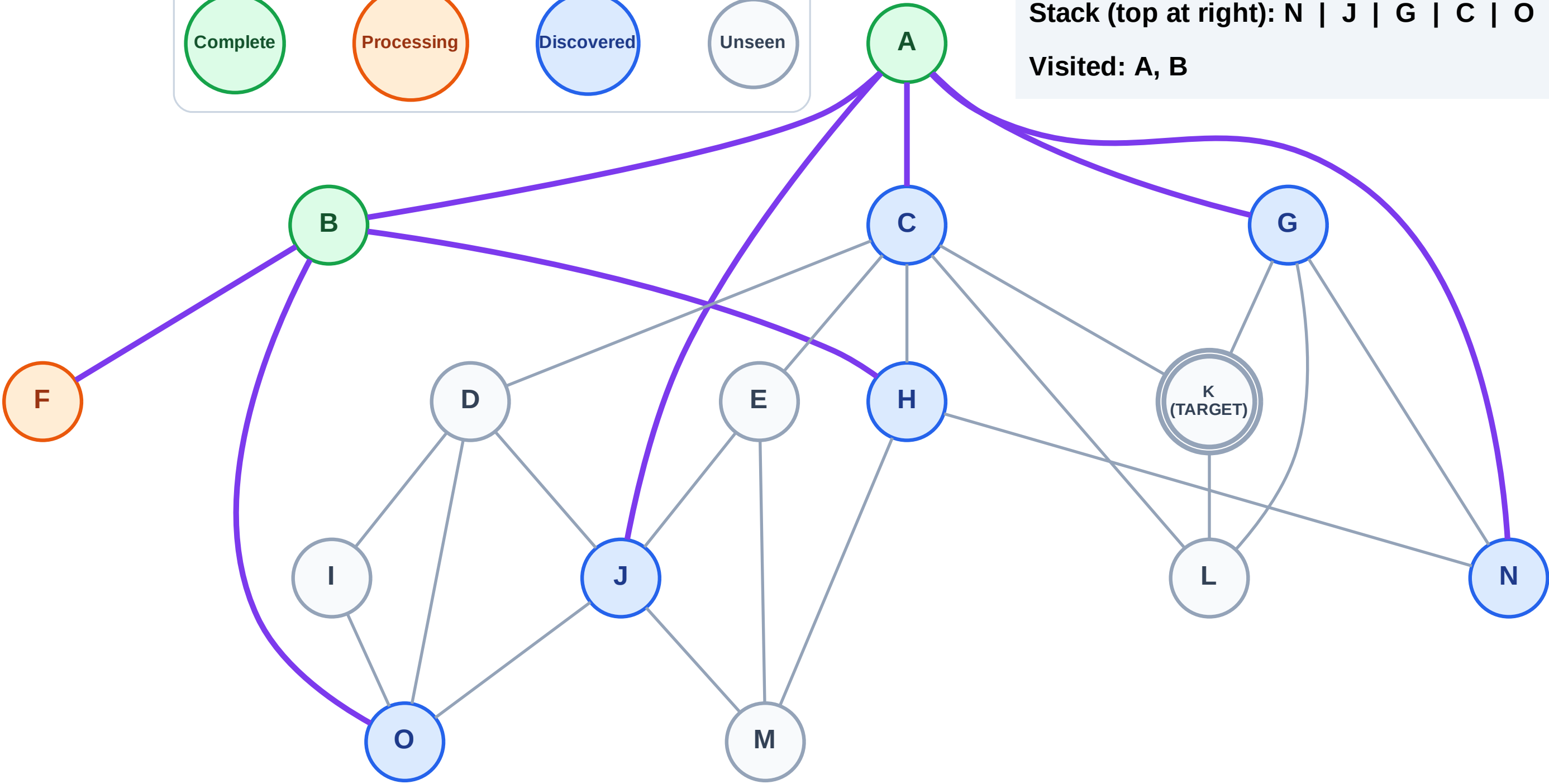
Processing

Discovered

Unseen

Stack (top at right): N | J | G | C | O | H

Visited: A, B



DFS Traversal

Step 7: No new neighbors from F

Legend

Complete

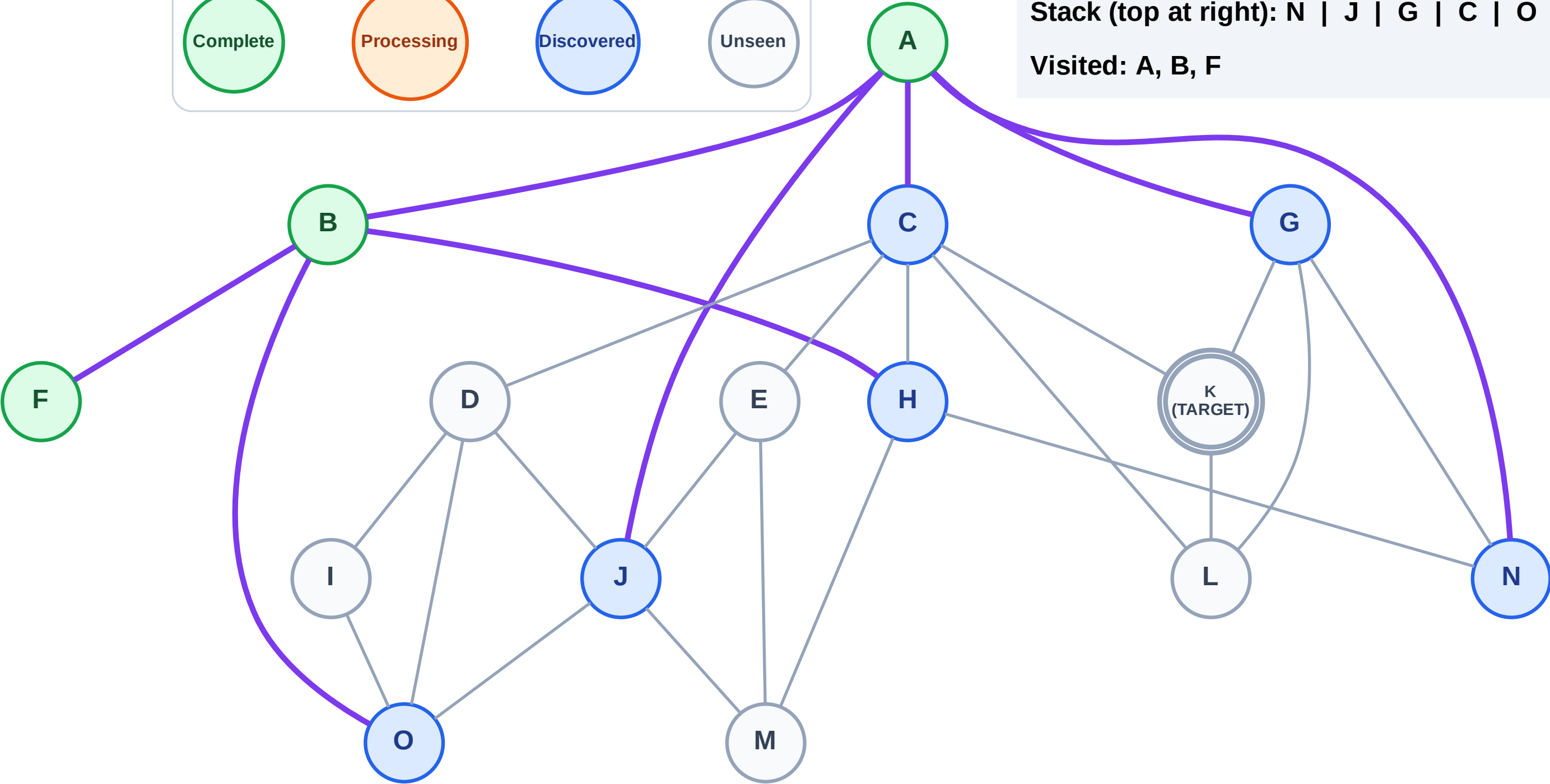
Processing

Discovered

Unseen

Stack (top at right): N | J | G | C | O | H

Visited: A, B, F



DFS Traversal

Step 8: Process node H

Legend

Complete

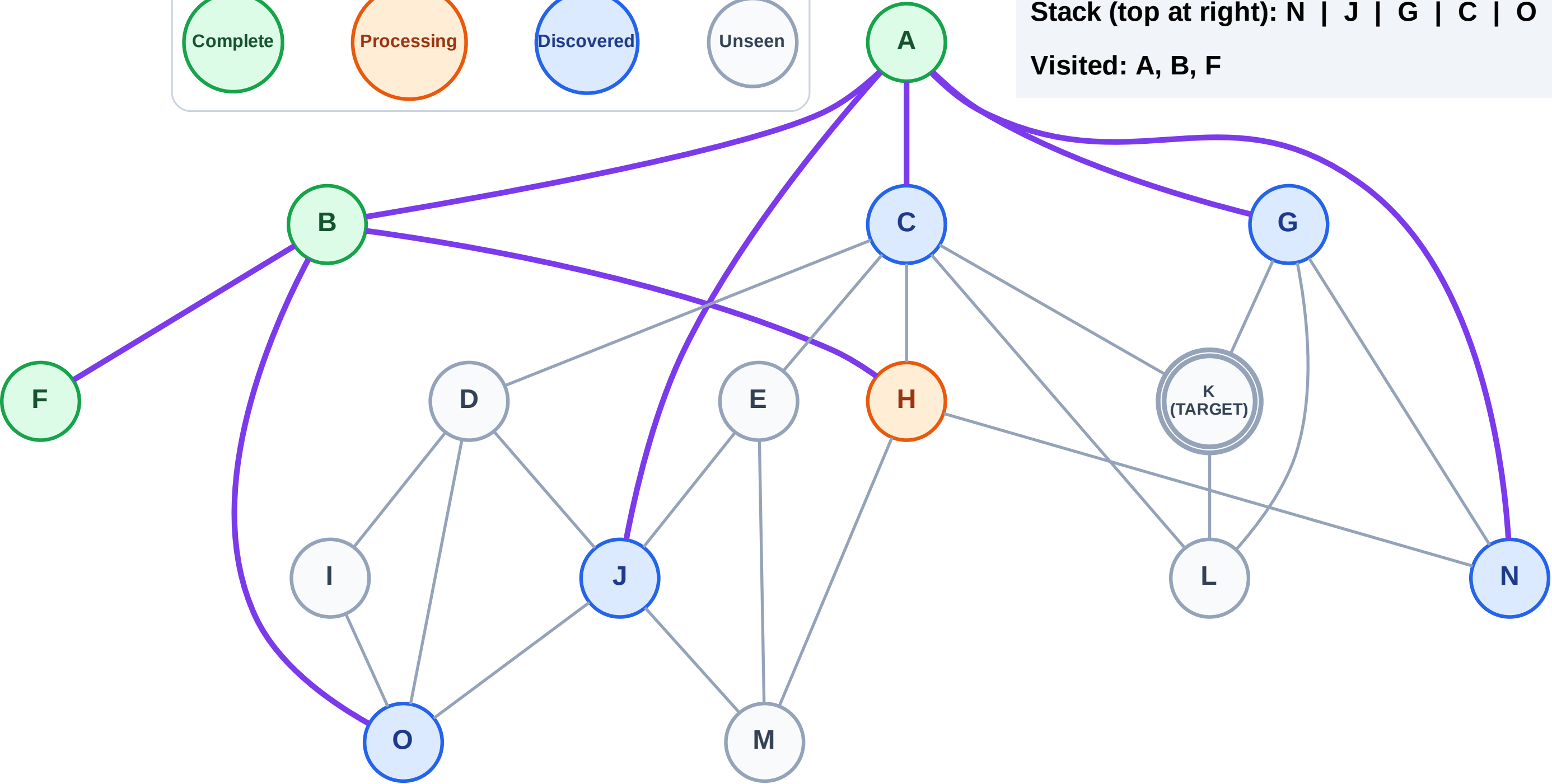
Processing

Discovered

Unseen

Stack (top at right): N | J | G | C | O

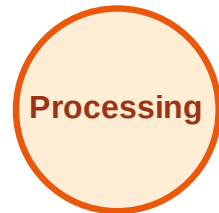
Visited: A, B, F



DFS Traversal

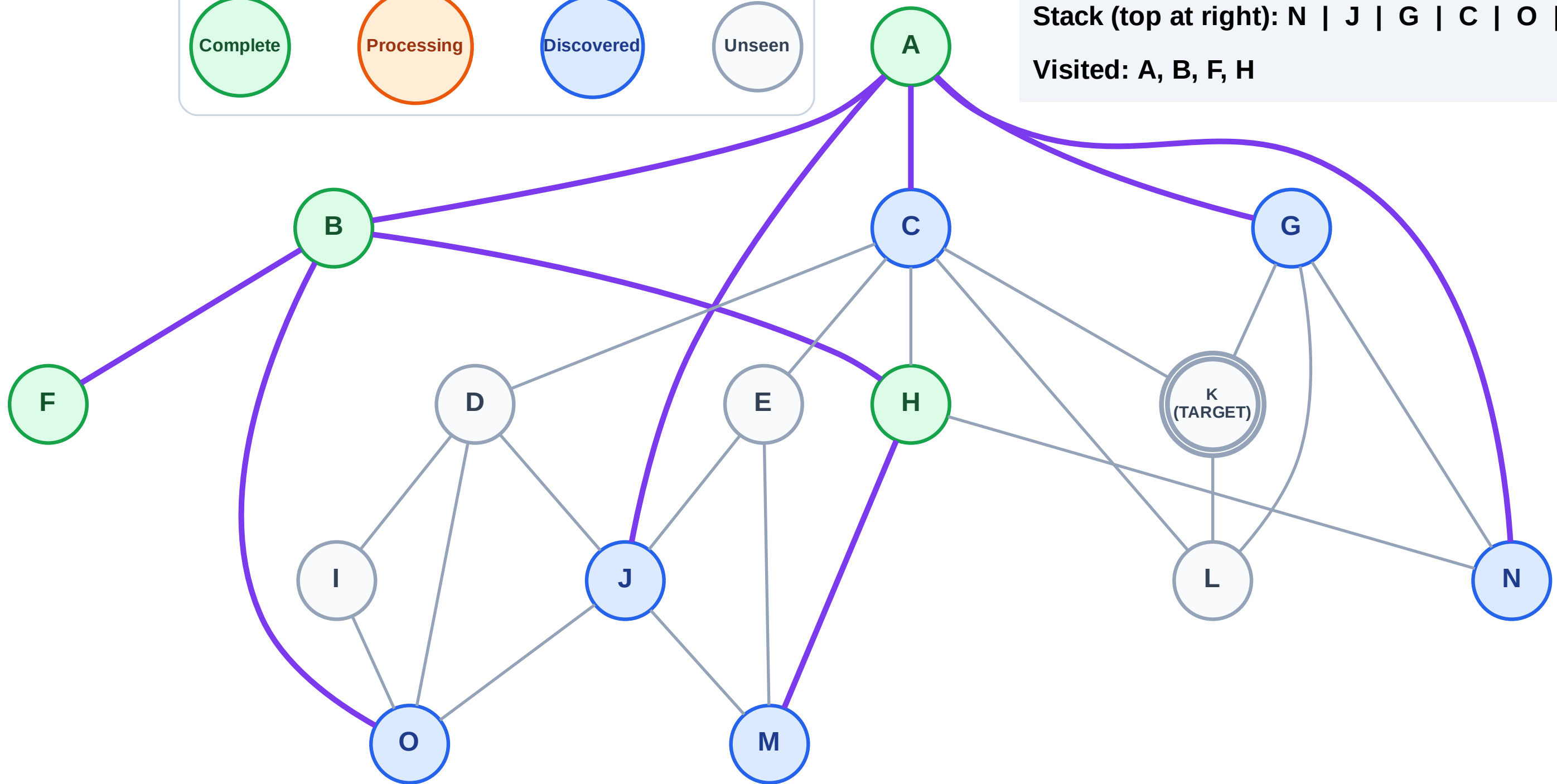
Step 9: Discover M from H

Legend



Stack (top at right): N | J | G | C | O | M

Visited: A, B, F, H



DFS Traversal

Step 10: Process node M

Legend

Complete

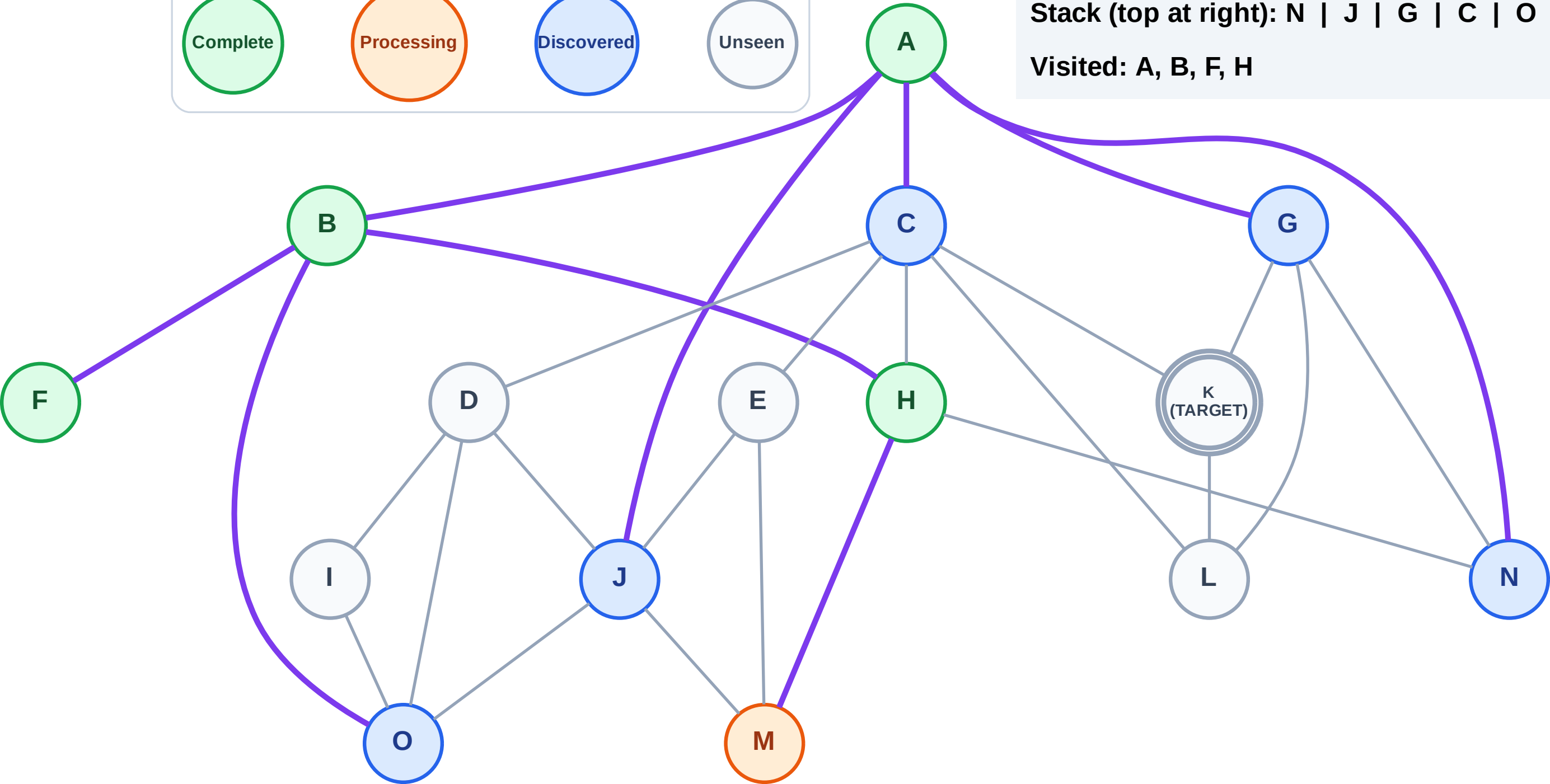
Processing

Discovered

Unseen

Stack (top at right): N | J | G | C | O

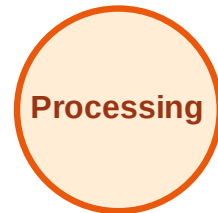
Visited: A, B, F, H



DFS Traversal

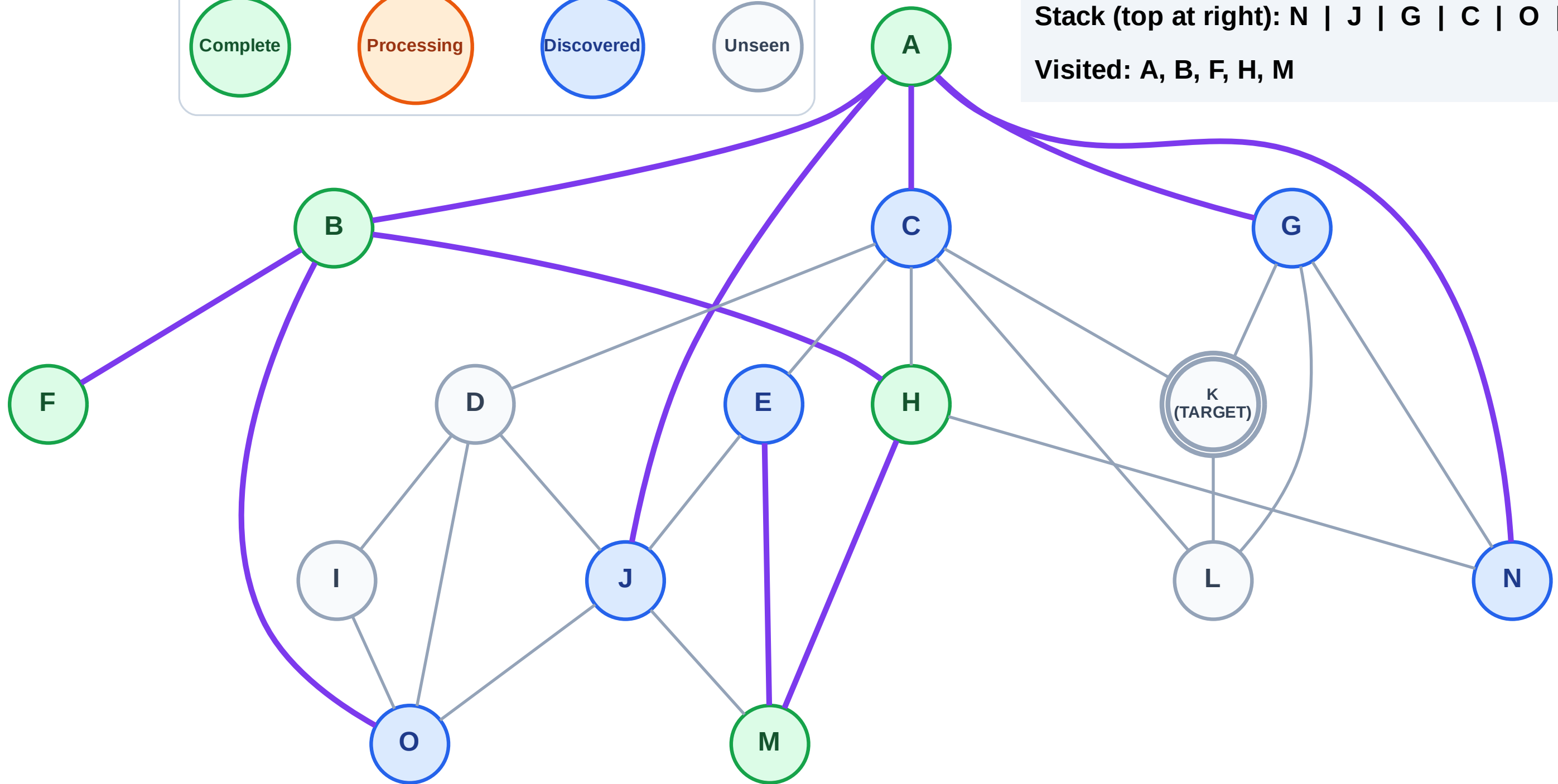
Step 11: Discover E from M

Legend



Stack (top at right): N | J | G | C | O | E

Visited: A, B, F, H, M



DFS Traversal

Step 12: Process node E

Legend

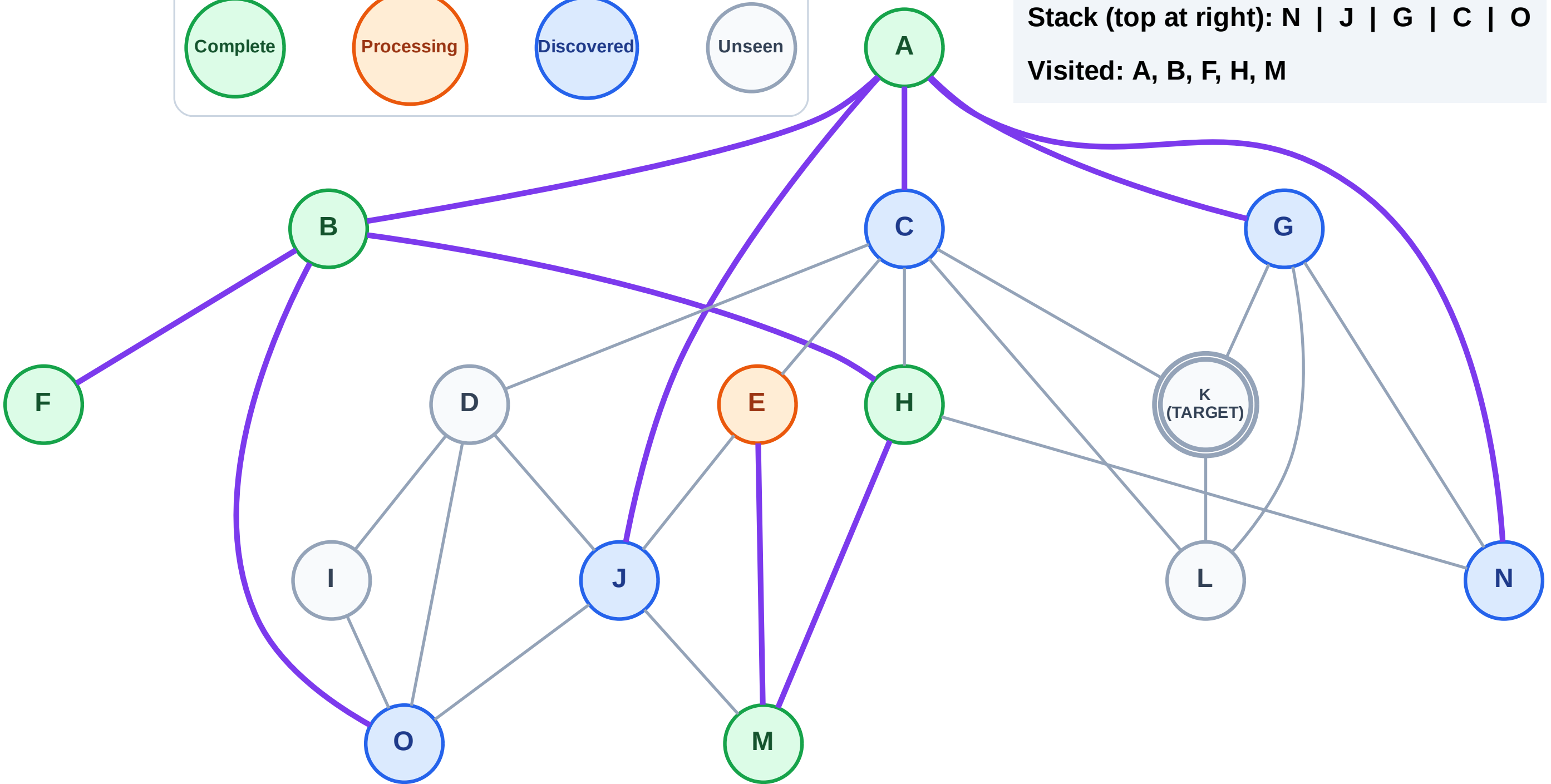
Complete

Processing

Discovered

Unseen

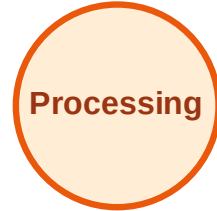
Stack (top at right): N | J | G | C | O
Visited: A, B, F, H, M



DFS Traversal

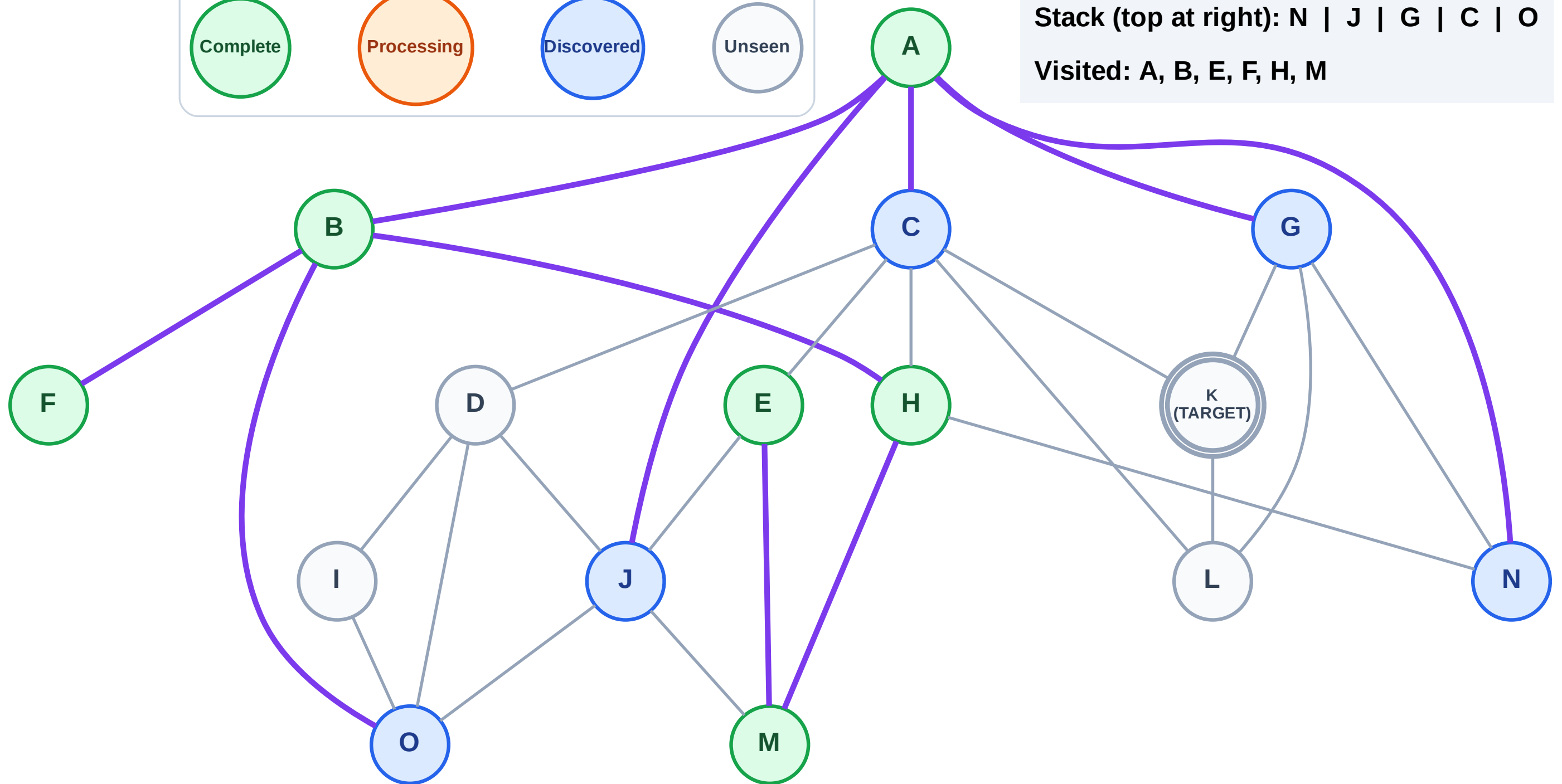
Step 13: No new neighbors from E

Legend



Stack (top at right): N | J | G | C | O

Visited: A, B, E, F, H, M



DFS Traversal

Step 14: Process node O

Legend

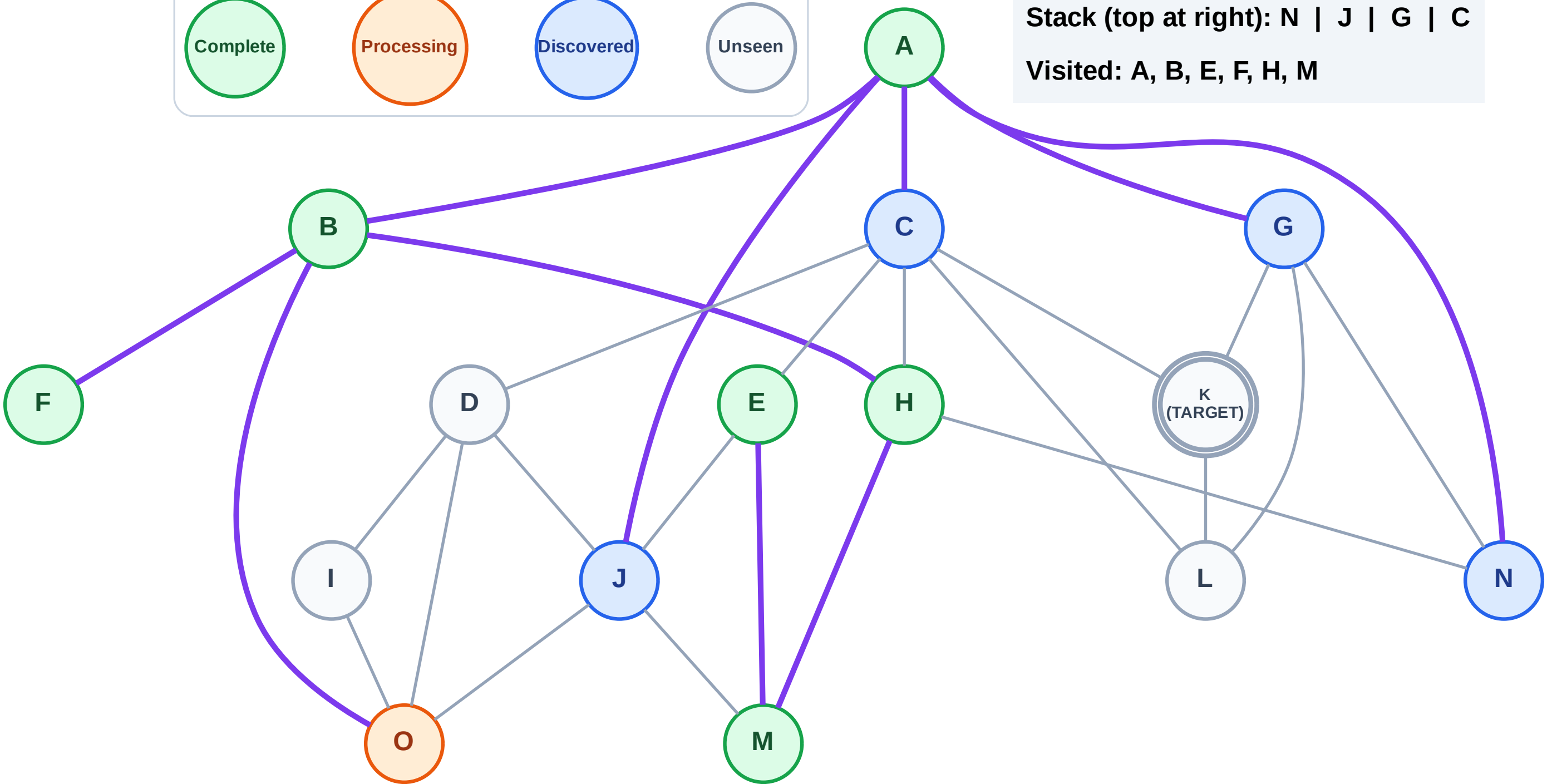
Complete

Processing

Discovered

Unseen

Stack (top at right): N | J | G | C
Visited: A, B, E, F, H, M



DFS Traversal

Step 15: Discover I, D from O

Legend

Complete

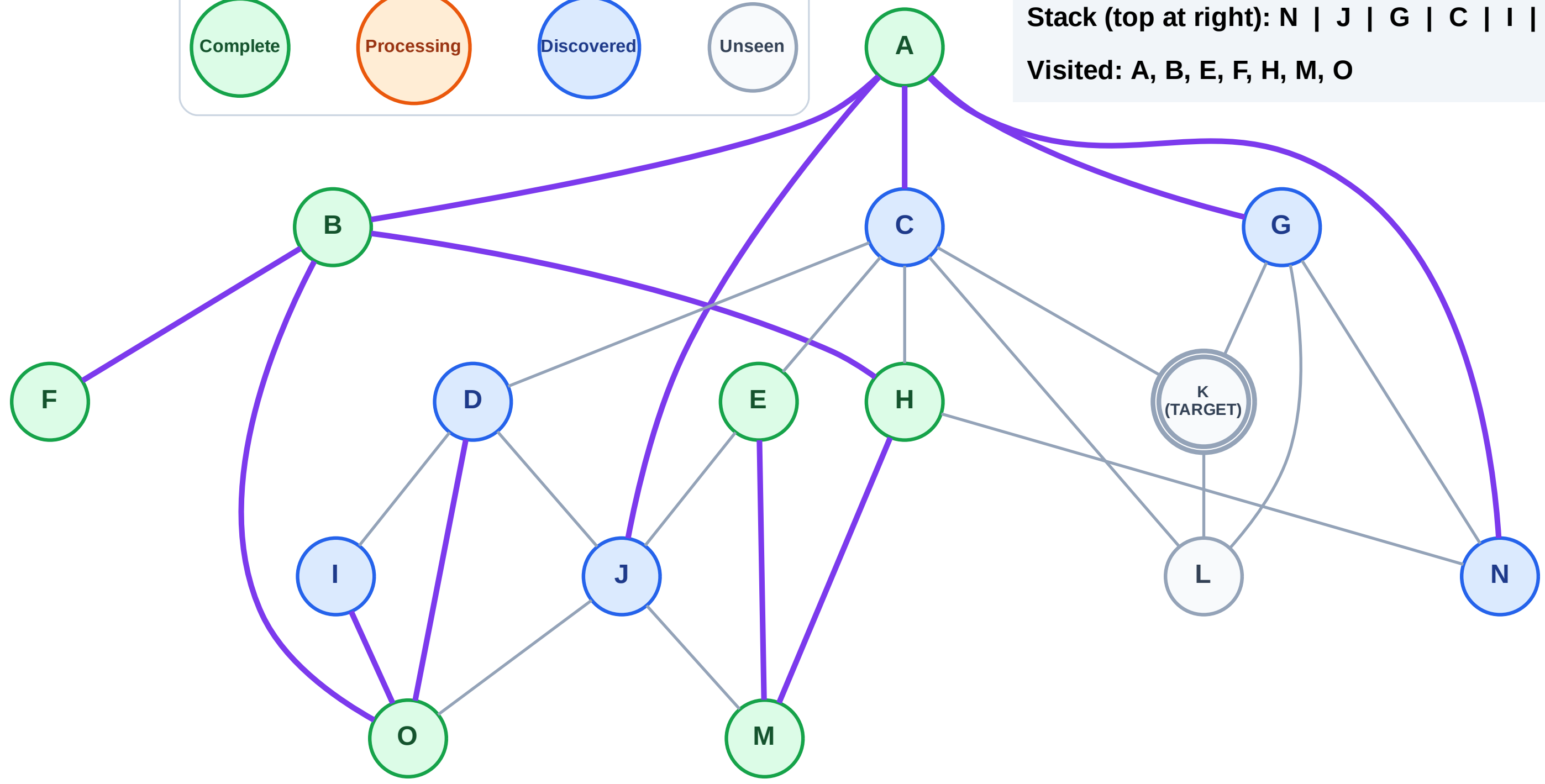
Processing

Discovered

Unseen

Stack (top at right): N | J | G | C | I | D

Visited: A, B, E, F, H, M, O



DFS Traversal

Step 16: Process node D

Legend



Complete



Processing



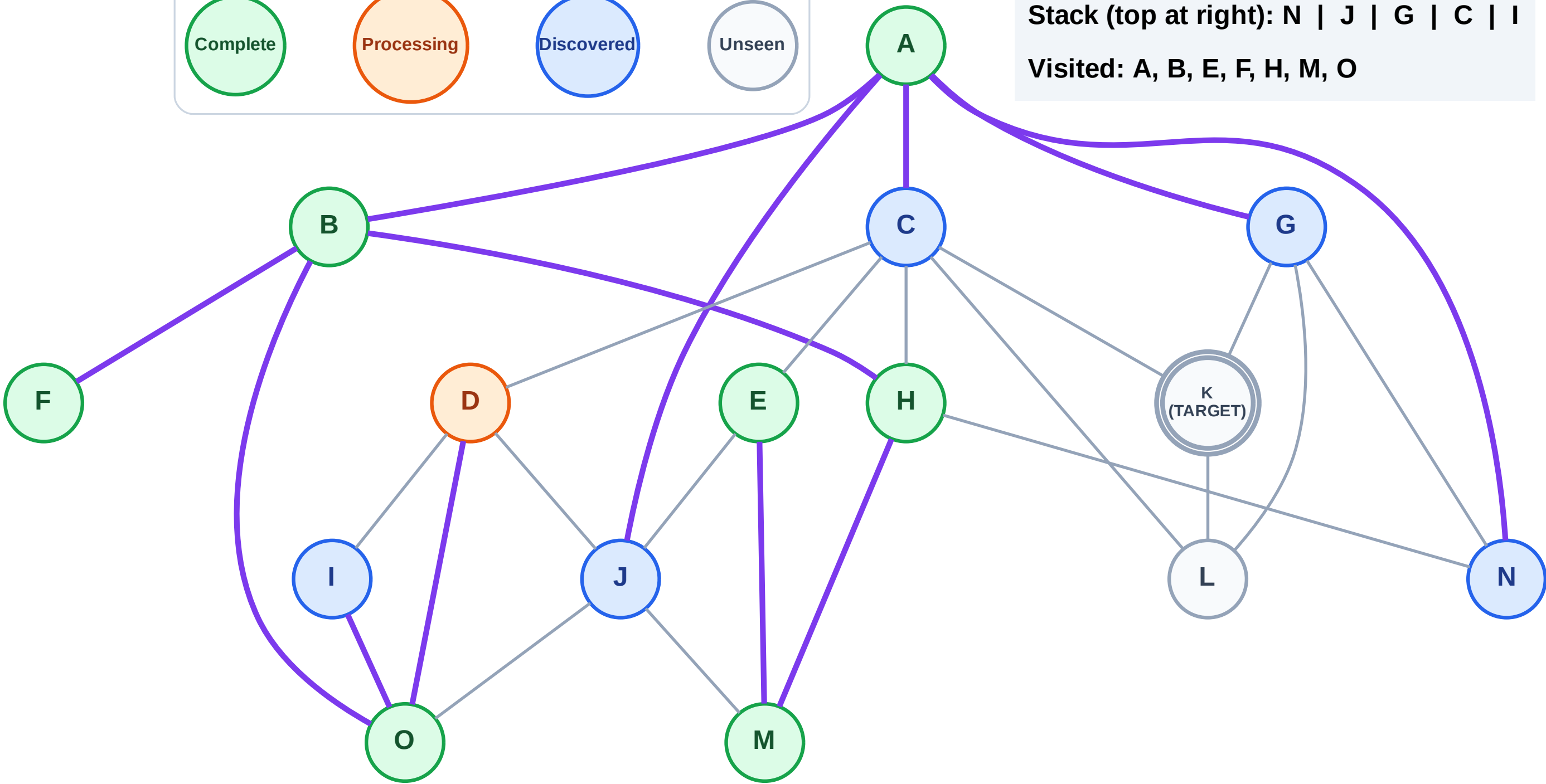
Discovered



Unseen

Stack (top at right): N | J | G | C | I

Visited: A, B, E, F, H, M, O



DFS Traversal

Step 17: No new neighbors from D

Legend



Complete



Processing



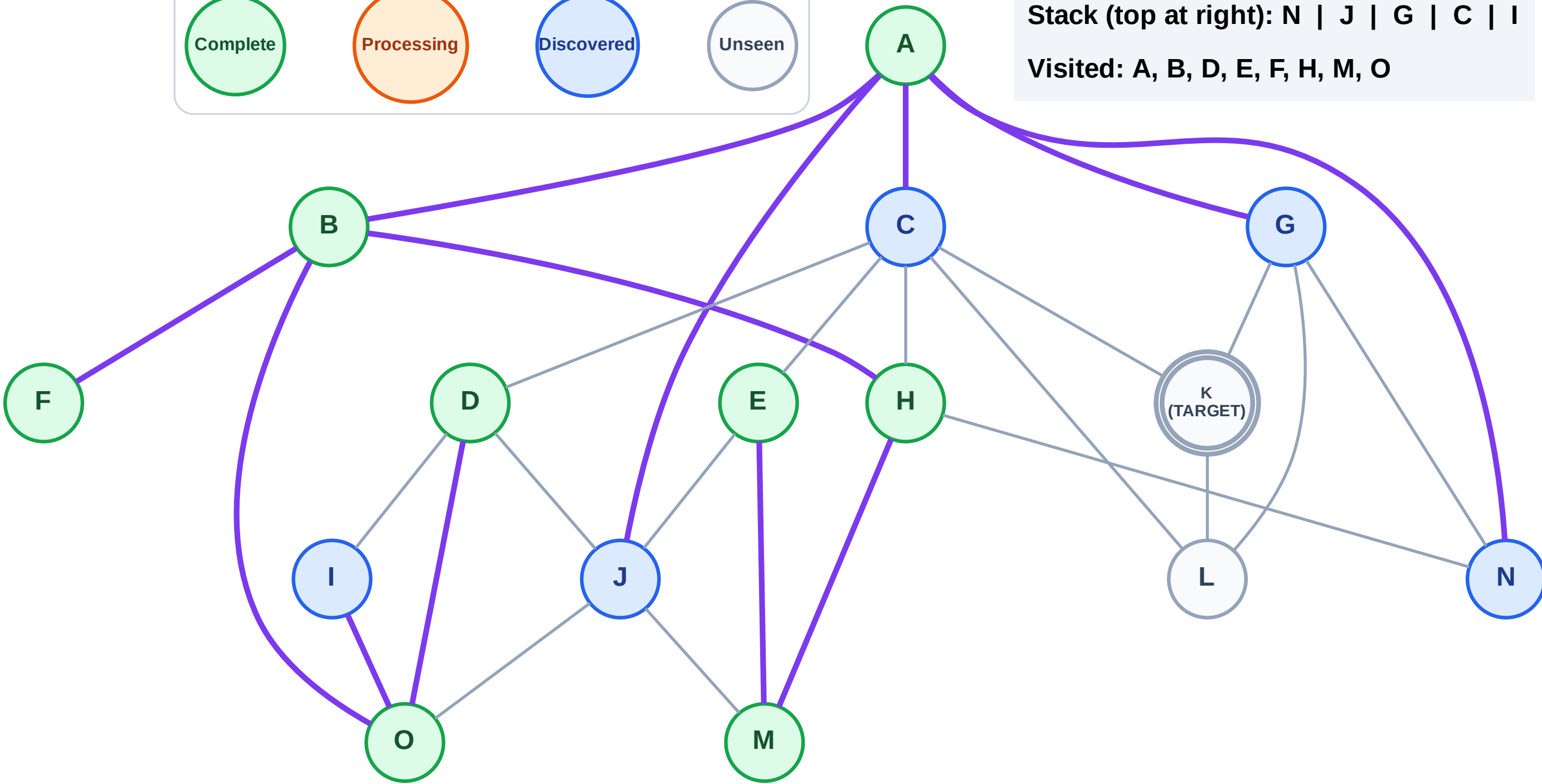
Discovered



Unseen

Stack (top at right): N | J | G | C | I

Visited: A, B, D, E, F, H, M, O



DFS Traversal

Step 18: Process node I

Legend

Complete

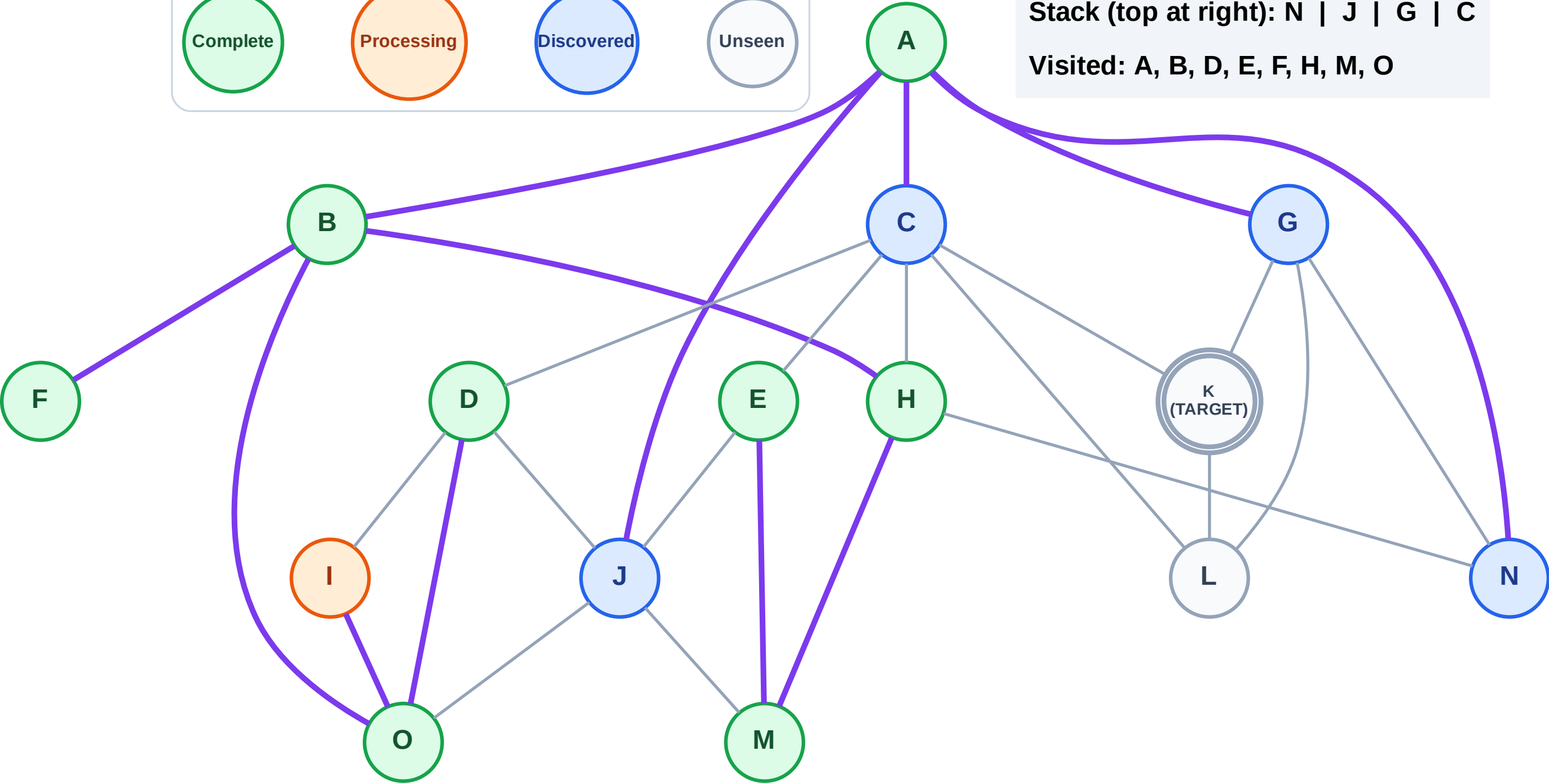
Processing

Discovered

Unseen

Stack (top at right): N | J | G | C

Visited: A, B, D, E, F, H, M, O



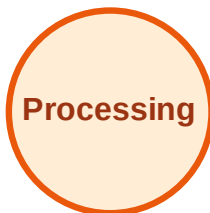
DFS Traversal

Step 19: No new neighbors from I

Legend



Complete



Processing



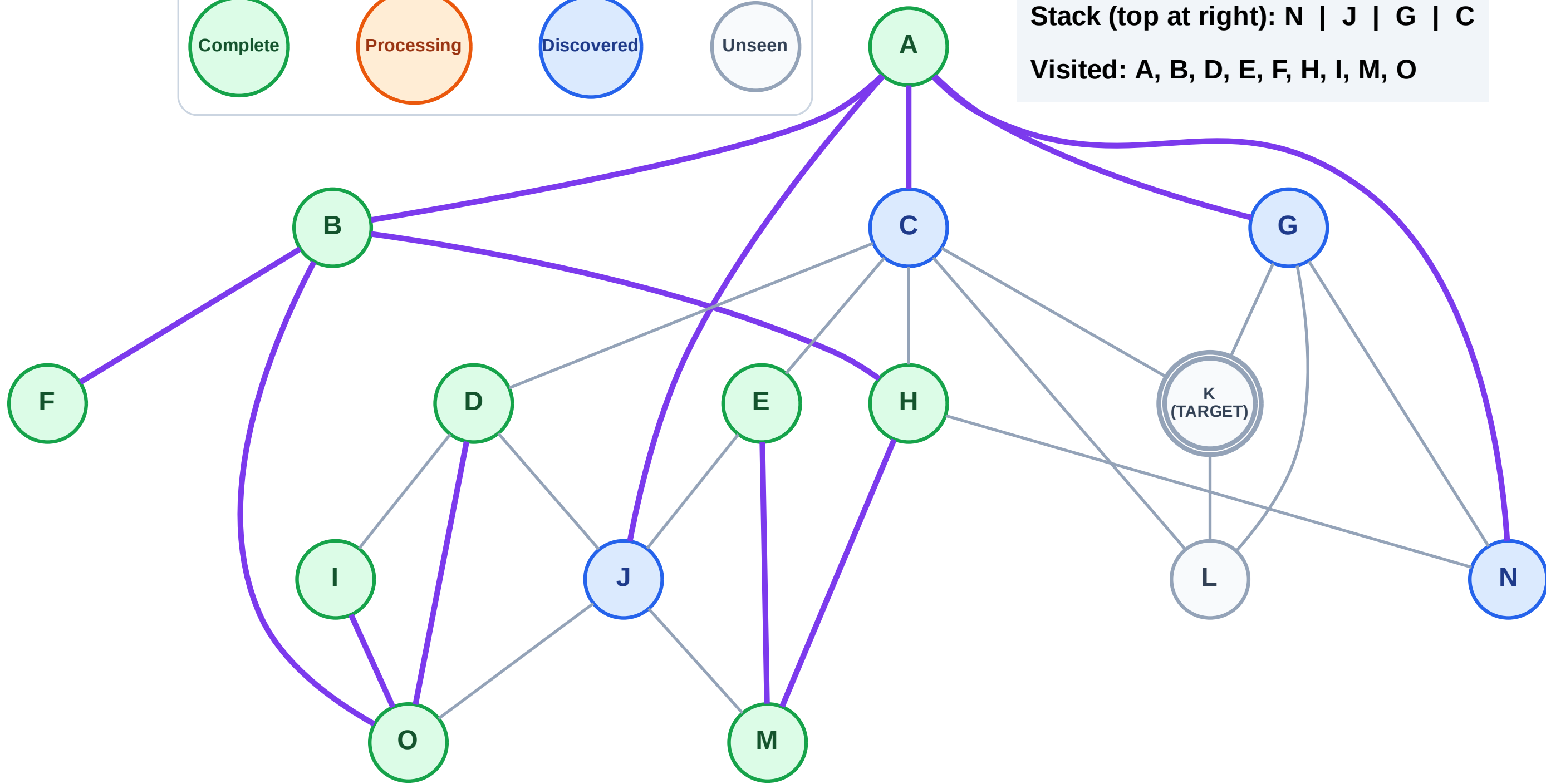
Discovered



Unseen

Stack (top at right): N | J | G | C

Visited: A, B, D, E, F, H, I, M, O



DFS Traversal

Step 20: Process node C

Legend

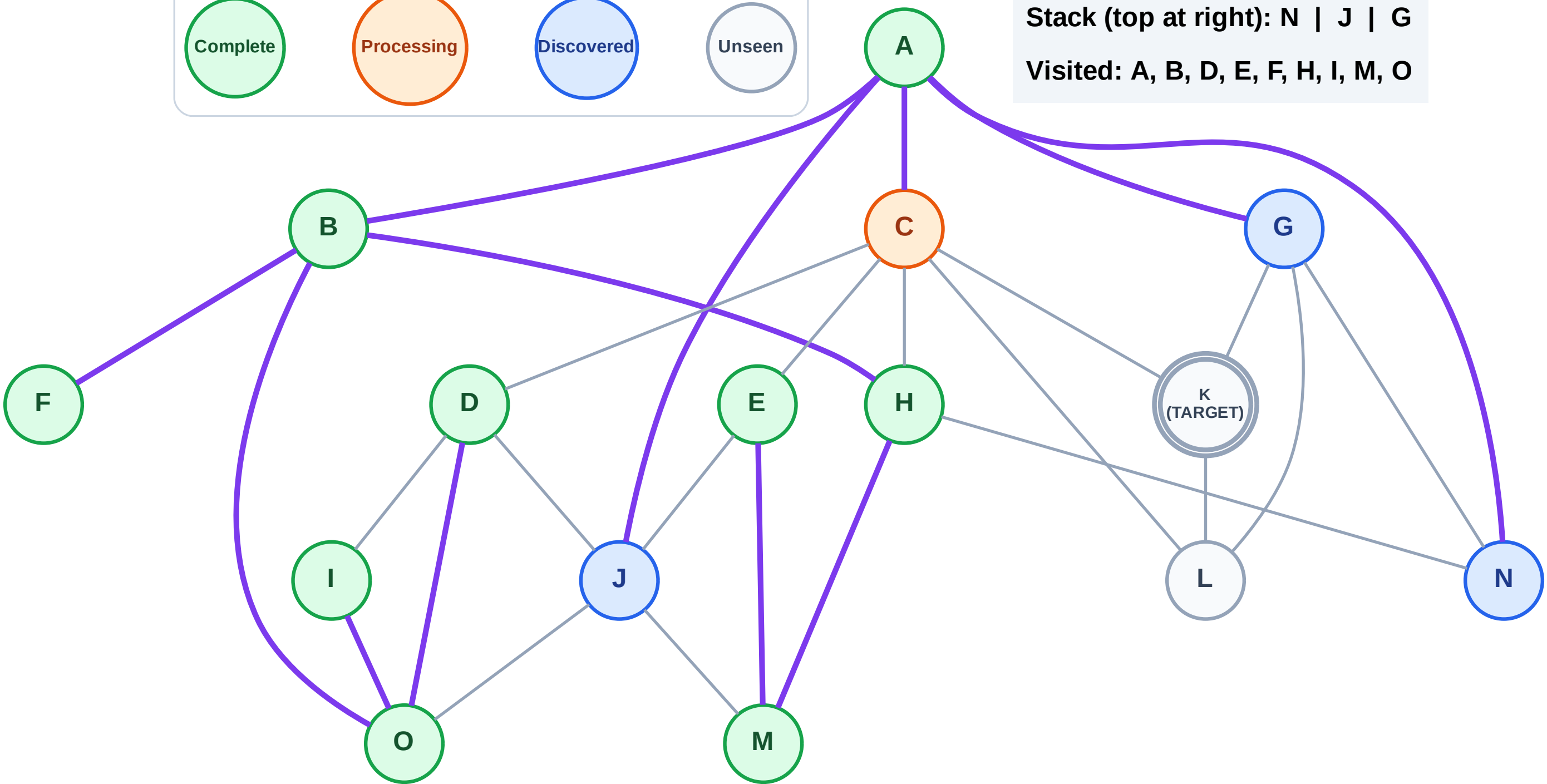
Complete

Processing

Discovered

Unseen

Stack (top at right): N | J | G
Visited: A, B, D, E, F, H, I, M, O



DFS Traversal

Step 21: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → C → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): N | J | G | L
Visited: A, B, C, D, E, F, H, I, K, M, O

